

EE4C10

Analog Circuit Design Fundamentals (References & Temp Sensors)

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Electronic Instrumentation Laboratory (EI)

Welcome to this track!

- Focus: CMOS **references**
- In this track you will learn about
 - The temperature dependency of CMOS devices (BJTs, MOSFETs, resistors)
 - Basic design of bandgap references
 - Basic design of constant-gm biasing circuits
 - The tolerances (spread) of CMOS devices

Study material

- Main:
 - Chapter 11, *Bandgap references* in B. Razavi's "Design of Analog CMOS Integrated circuits".
- For your interest:
 - Chapter 3, *Ratiometric temperature measurement using bipolar transistors* in M. Pertijs's "Precision Temperature Sensors in CMOS Technology".

Simulator and track assignment

- Software: LTspice XVII
- Library file
 - log018.l & log018_mod.l
- Examples can be found in Brightspace
- Assignment: bandgap references
- Report in .pdf format, with screenshots and explanations.



On-chip References

- Stable circuit operation requires stable **on-chip** references
- We will study two types
 - Constant-gm bias circuits
 - Bandgap references

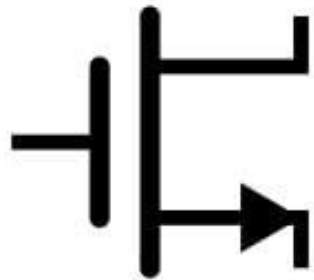
Recap: MOSFET model

- In strong-inversion region (saturation)

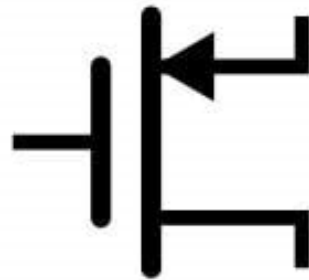
$$I_D \approx \frac{\mu_n C_{ox}}{2} \frac{W}{L} (V_{GS} - V_{th})^2$$

But μ_n and V_{th} are temperature dependent!

NMOS

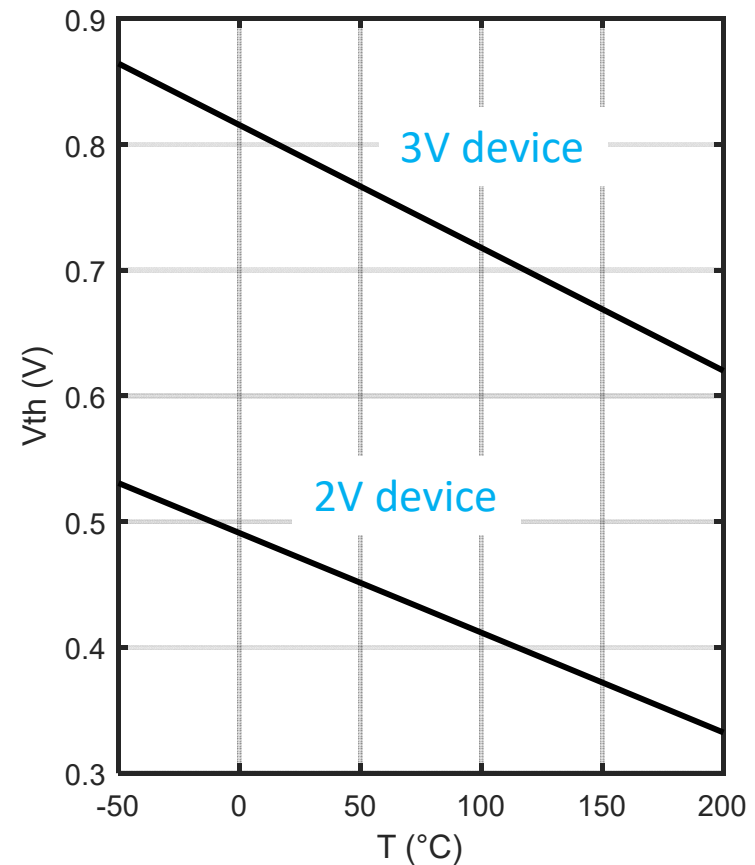
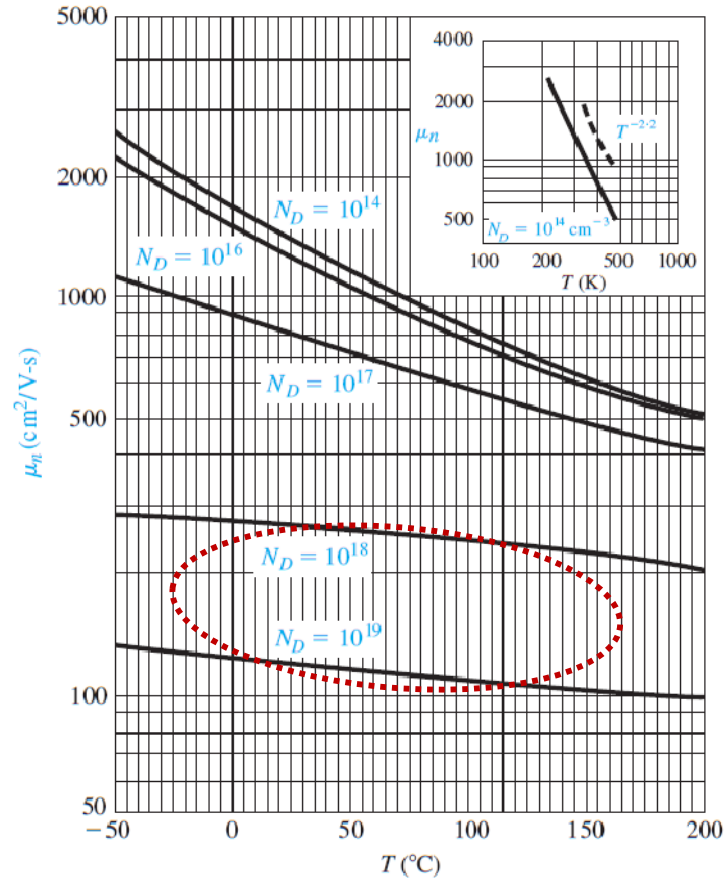


PMOS



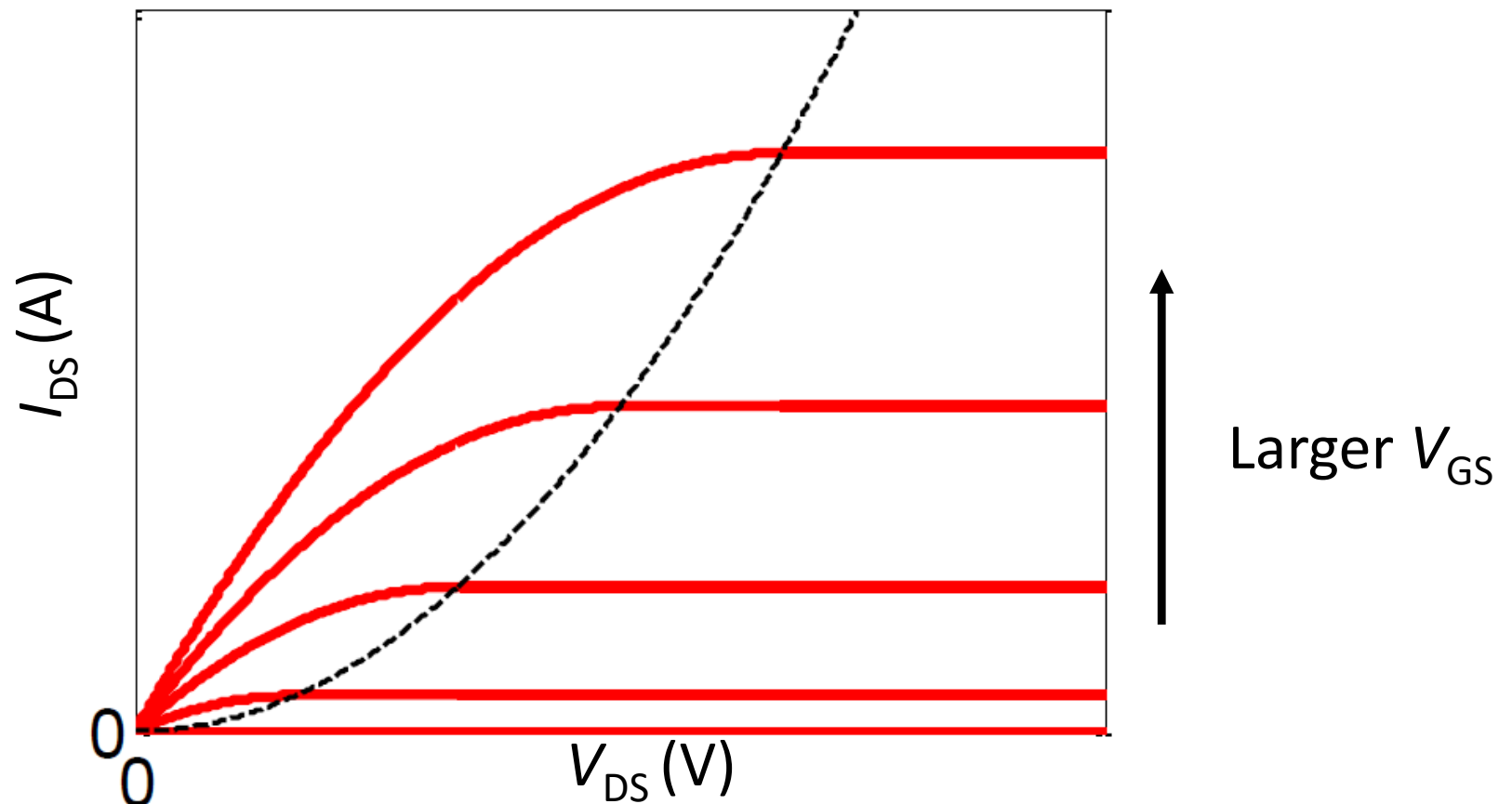
MOSFET: temperature dependency

- Temperature dependence of μ_n and V_{th} (NMOS)



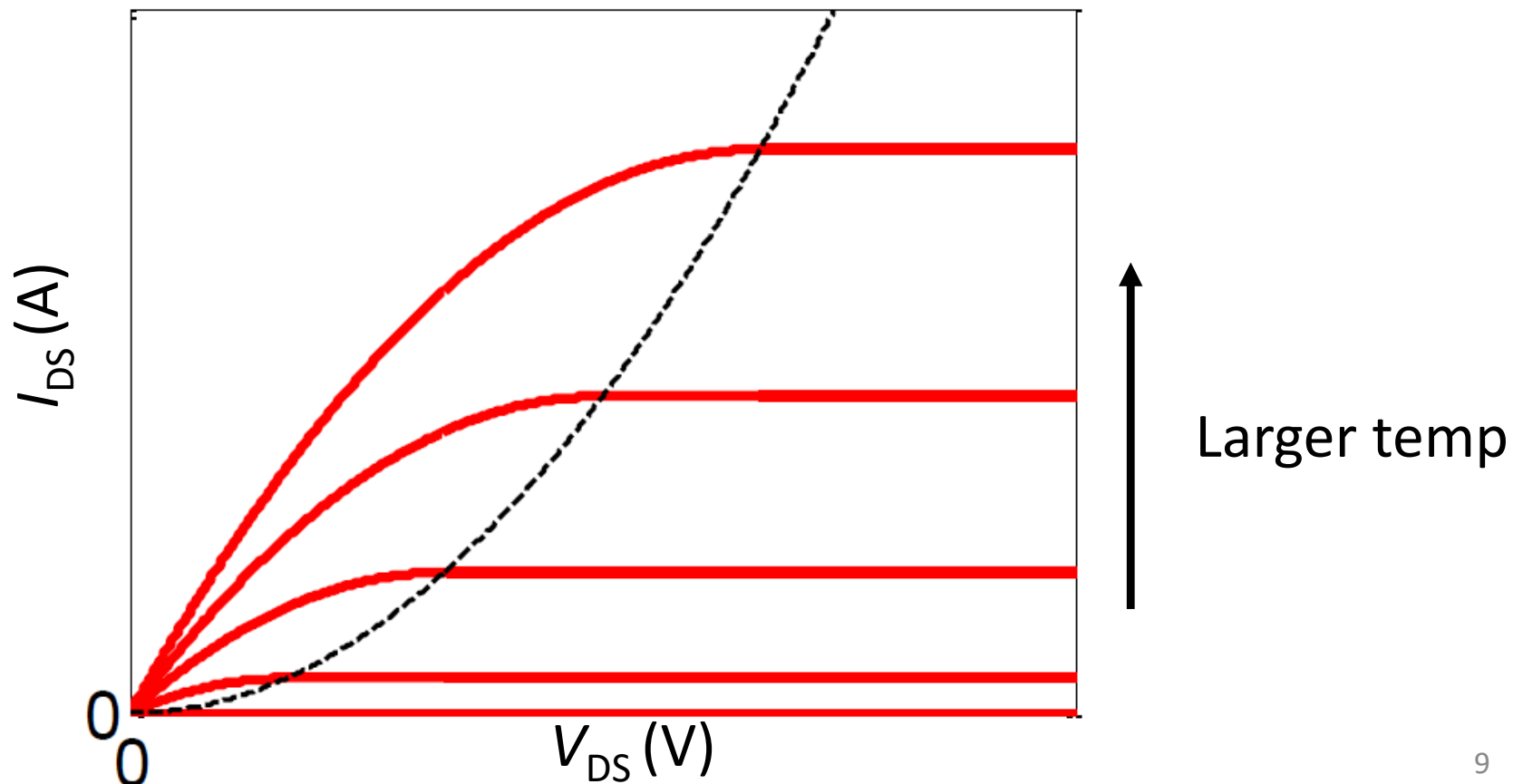
Recap: MOSFET characteristic

- Voltage controlled current source
- I_{DS} mainly determined by V_{GS} .



MOSFET Temp. characteristic

- $I_D \approx \frac{\mu_n C_{ox}}{2} \frac{W}{L} (V_{GS} - V_{th})^2$.
- Temp $\uparrow$, $\mu_n \downarrow$, $(V_{GS} - V_{th})^2 \uparrow \uparrow \Rightarrow I_{DS} \uparrow$.



BJT model

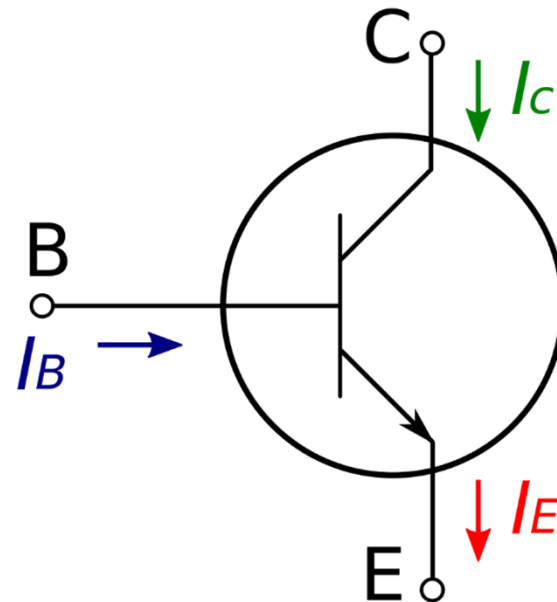
- Often modelled as a current-controlled current source.

$$I_C = \beta_F \cdot I_B \quad (\beta_F \gg 1)$$

- More accurately, in the forward-active region:

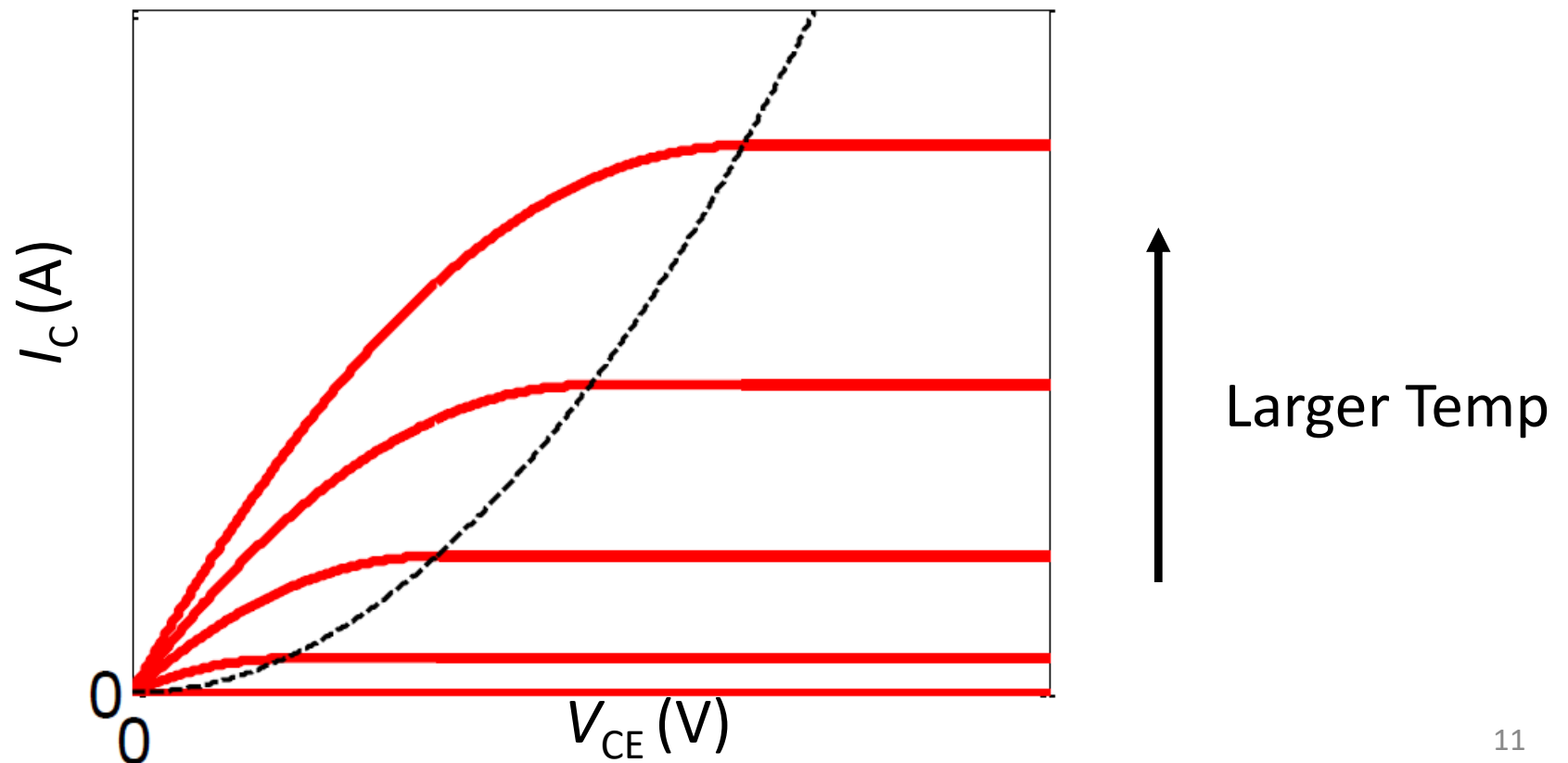
$$I_C \approx I_S \cdot e^{\frac{V_{BE}}{V_T}}$$

$$\text{where } V_T = kT/q$$



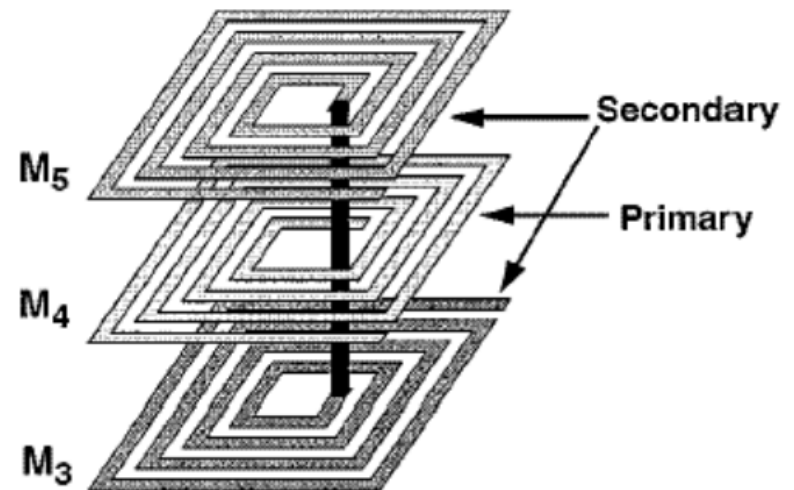
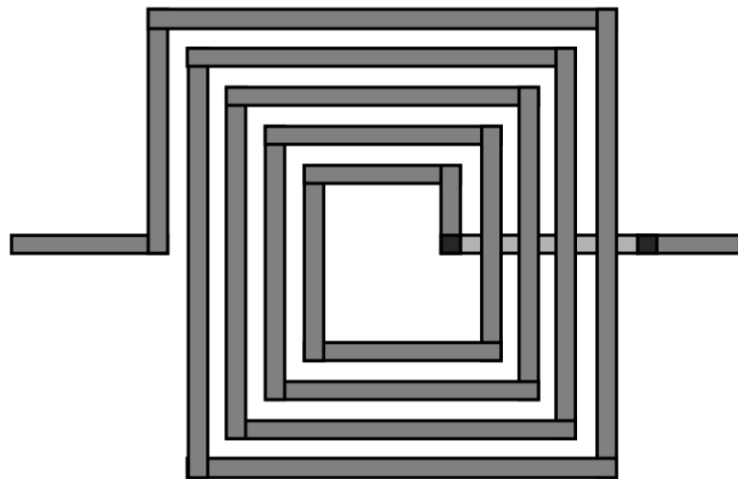
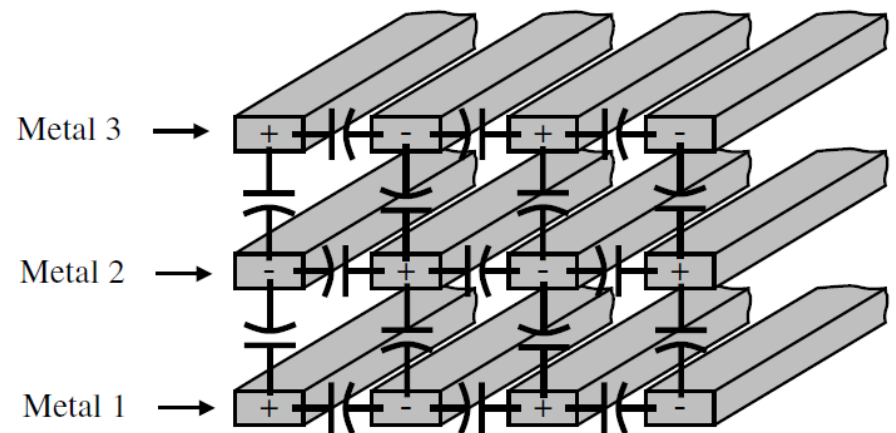
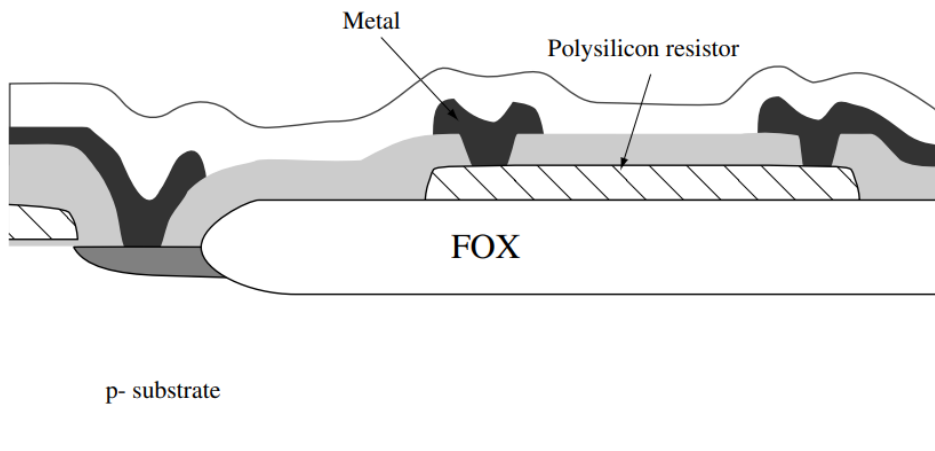
BJT Temp. characteristic

- $I_C \approx I_S \cdot e^{\frac{V_{BE}}{V_T}}$,
- Temp $\uparrow$, $e^{\frac{V_{BE}}{V_T}} \downarrow$, $I_S \uparrow\uparrow \Rightarrow I_C \uparrow$.



Other components?

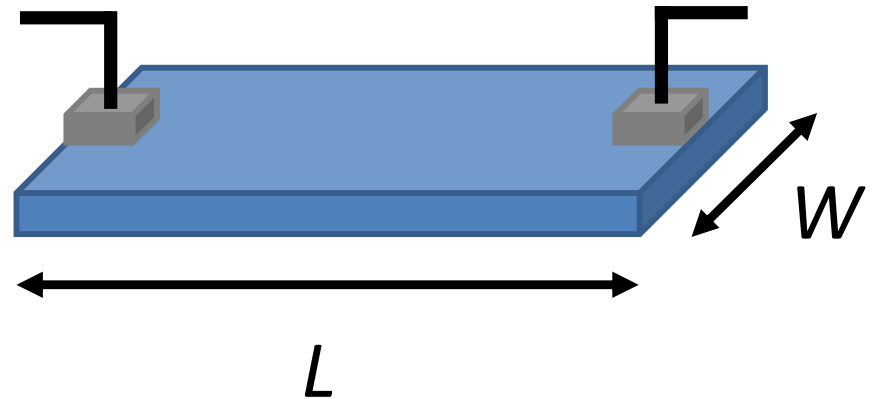
- Resistors, capacitors, inductors, transformers.



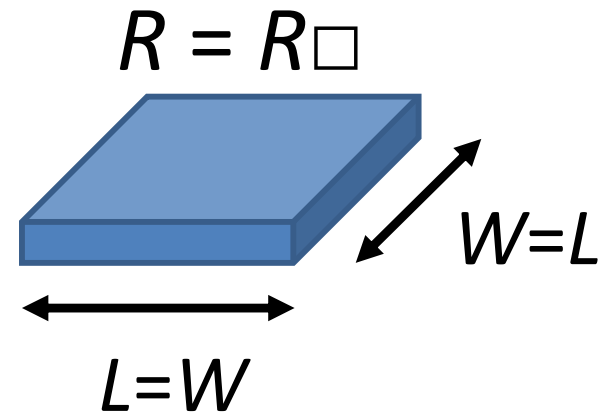
Resistors

- With a fixed thickness.

$$R = R_{\square} * (L/W)$$



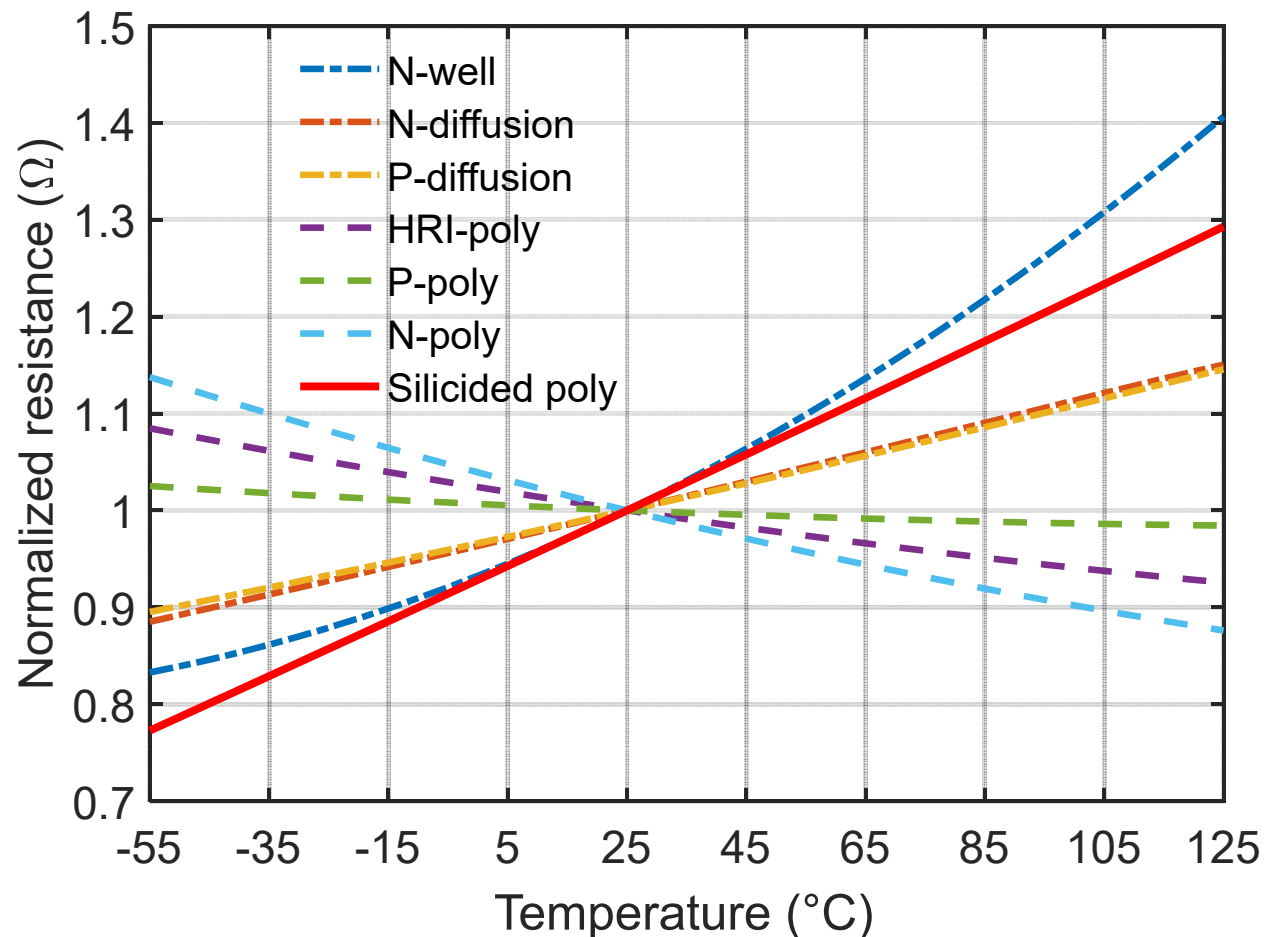
- 1st & 2nd order TC depends on resistor type (poly, diffusion, metal)



- $R_{\square} = R_{\square_0} * (1 + TC_1 * (T - T_0) + TC_2 * (T - T_0)^2)$

Resistors

- In 180nm CMOS, TCs range from -1500ppm/°C (n-poly) to 3000ppm/°C (N-well)

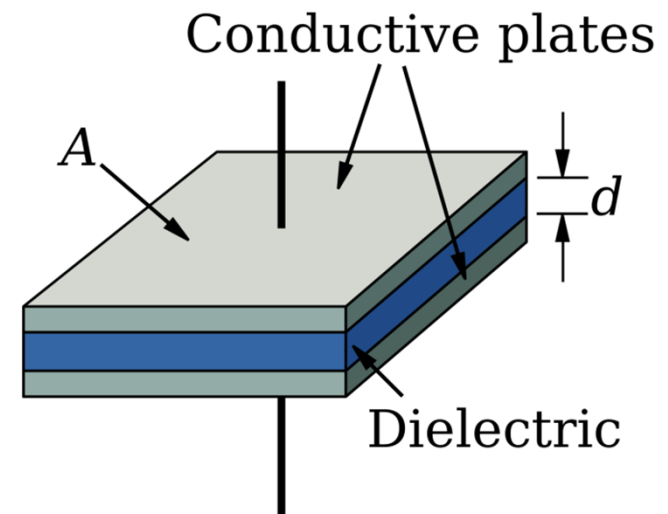
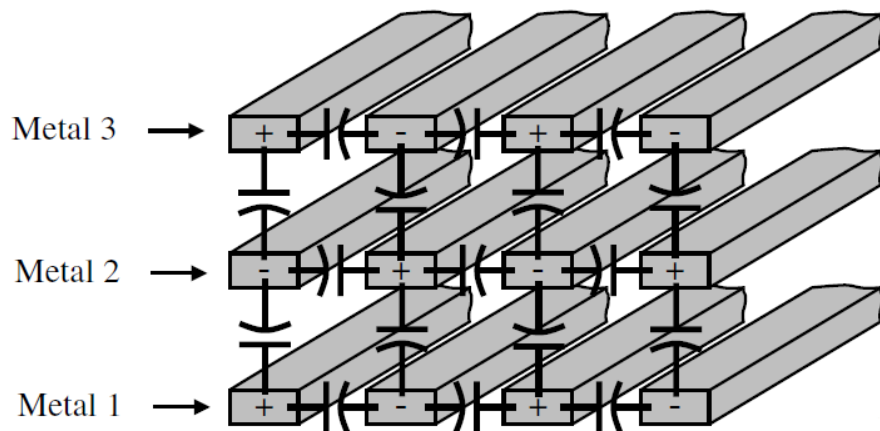


Capacitors

- Parallel plate capacitors:

$$C = \epsilon_0 \cdot A/d$$

- Very stable: TCs < 100ppm/°C



Constant-gm biasing

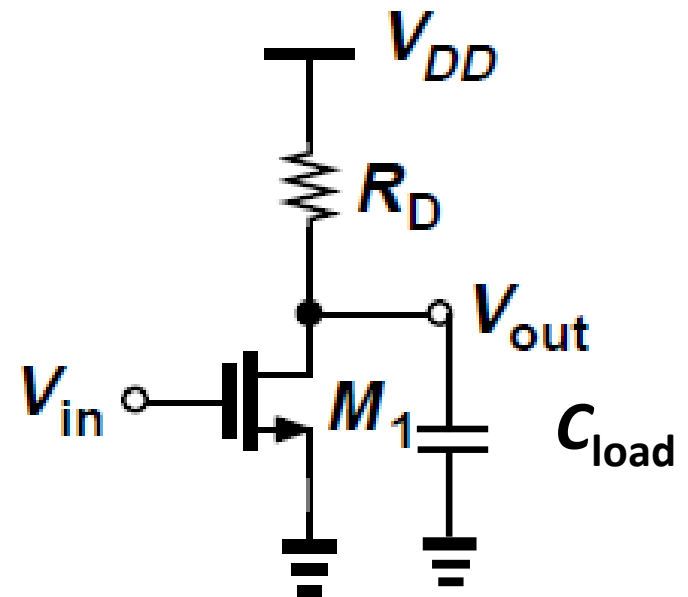
- Consider a simple single stage amplifier

$$F_{\text{unity}} = g_m / 2\pi C_{\text{load}}$$

- Constant g_{m1} over temperature

-> almost constant F_{unity}

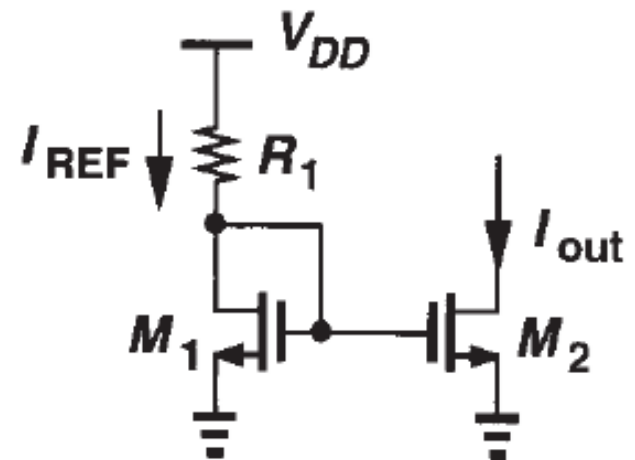
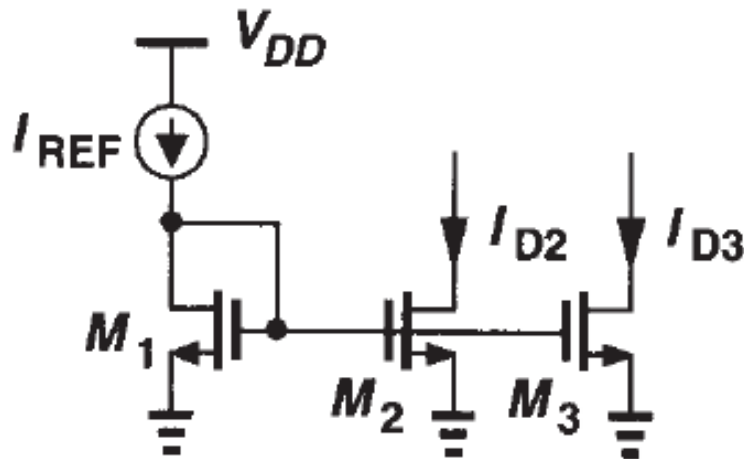
- Requires a constant-gm bias circuit??



The simplest biasing

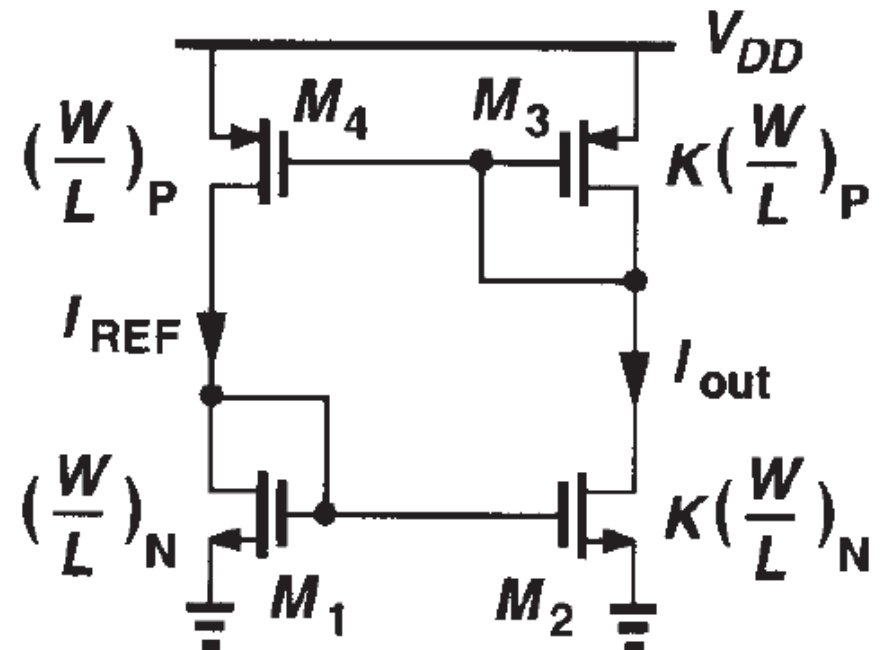
- No “golden” reference current available
- Poor-man’s circuit: sensitive to V_{DD}

- $$\Delta I_{out} = \frac{\Delta V_{DD}}{R_1 + 1/g_{m1}} \cdot \frac{(W/L)_2}{(W/L)_1}$$



Supply-independent current

- The circuit must somehow bias itself?
- Idea: Derive I_{REF} from I_{out}
- But what defines the current?



Supply-independent biasing

- Use a resistor to define the current
- Assume $\lambda=0$

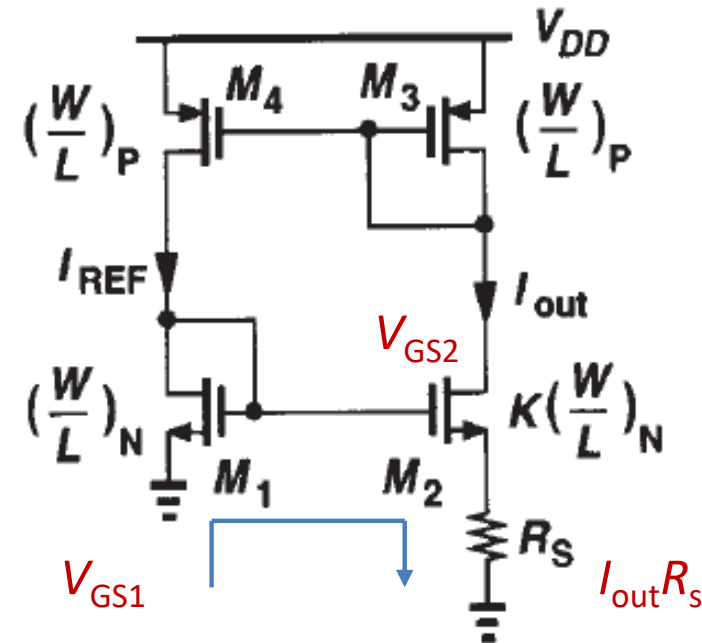
$$- I_{out} = I_{REF}$$

$$- V_{GS1} = V_{GS2} + I_{out} R_S$$

- Solving the equation leads to

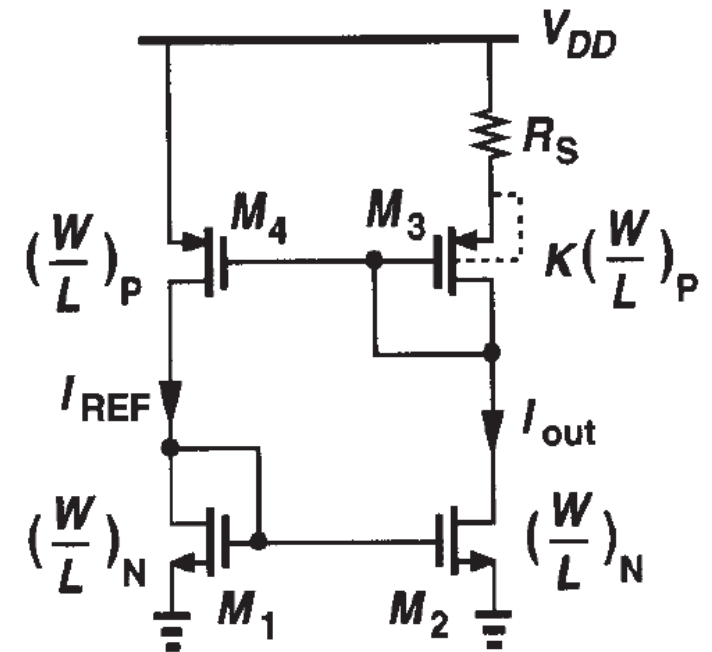
$$- I_{out} = \frac{2}{\mu_n C_{ox} (W/L)_N} \cdot \frac{1}{R_S^2} \left(1 - \frac{1}{\sqrt{K}} \right)^2$$

- Better, but note that μ_n is still a function of temperature



Supply-independent biasing

- To mitigate the body effect (back-gate effect) PMOS transistors are better
- Their back-gates can be tied to the source instead of Vdd

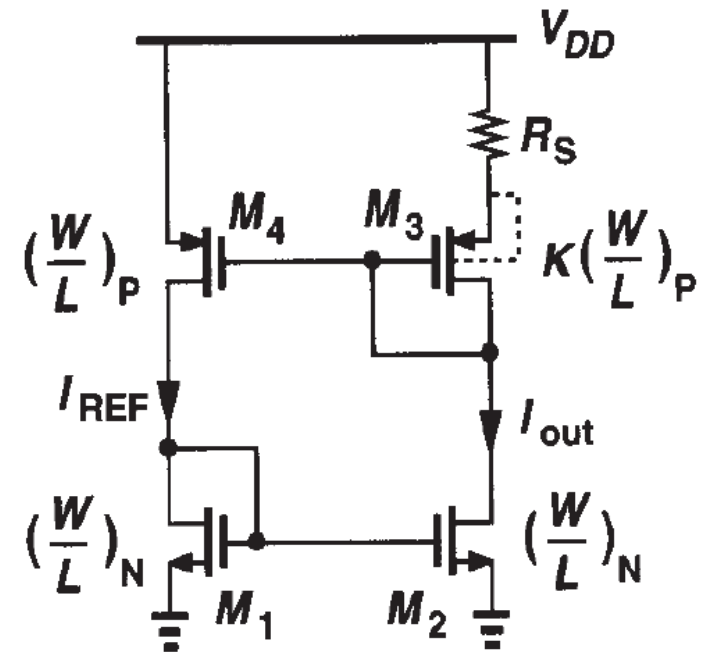


Constant-gm biasing!

- $$I_{out} = \frac{2}{\mu_n C_{ox} (W/L)_N} \cdot \frac{1}{R_S^2} \left(1 - \frac{1}{\sqrt{K}}\right)^2$$
- $$g_{m1} = \sqrt{2\mu_n C_{ox} \left(\frac{W}{L}\right)_N I_{D1}}$$

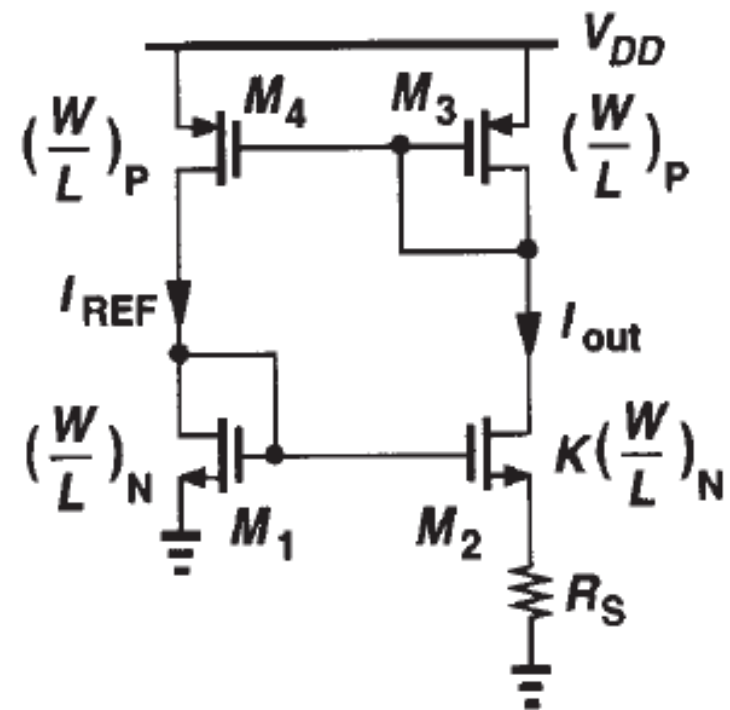
$$= \frac{2}{R_S} \left(1 - \frac{1}{\sqrt{K}}\right)$$

Only a function of R_S and K !



Start-up issue

- $V_{GS1} = V_{GS2} = 0$
- $V_{GS3} = V_{GS4} = 0$
- No current flowing through any device!



Start-up issue: solution

- Start-up sequence

$$V_{GS1} = V_{GS2} = V_{GS3} = V_{GS4} = 0$$

-> M5 switched on

-> M1 and M3 turned on

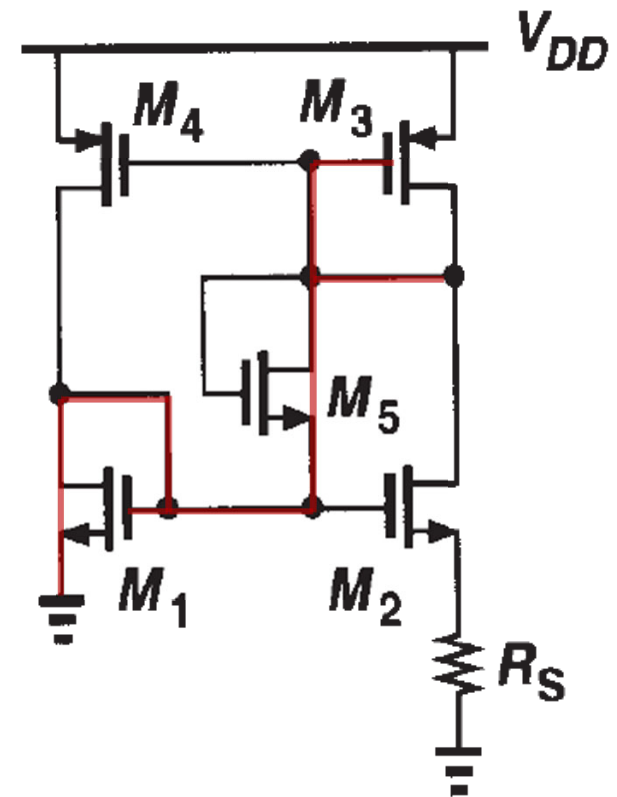
-> M2 and M4 turned on

-> M5 turned off

- Conditions:

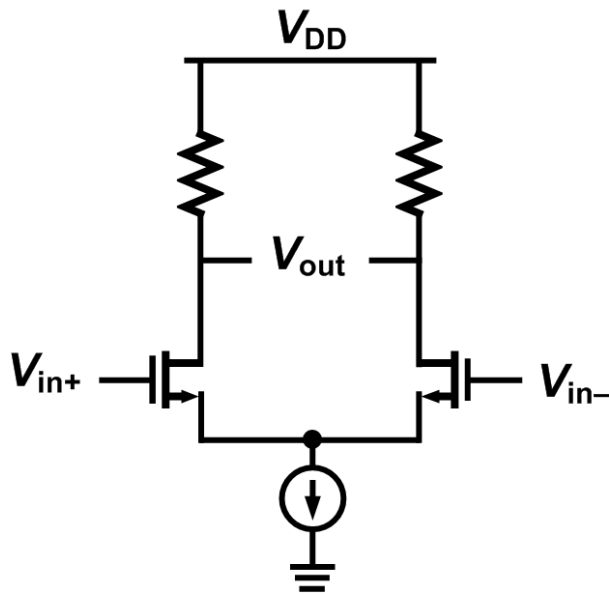
$$V_{TH1} + V_{TH5} + |V_{TH3}| < V_{DD}$$

$$V_{GS1} + V_{TH5} + |V_{GS3}| > V_{DD}$$

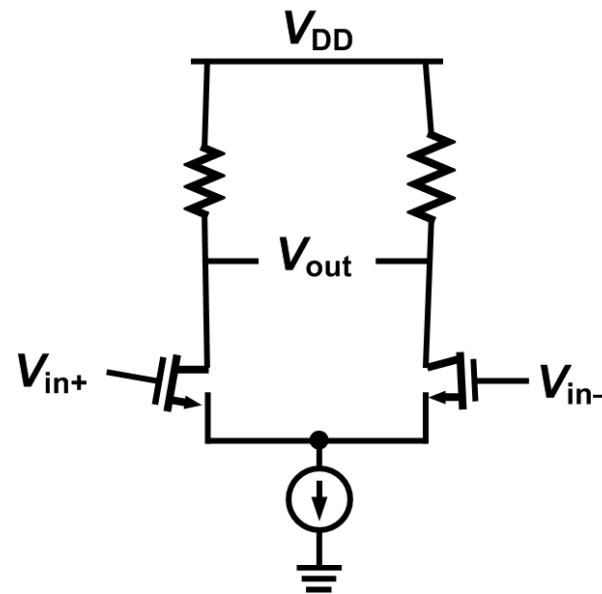


Spread and tolerance

- So far we have assumed perfectly accurate devices
- But in the real world all devices spread!



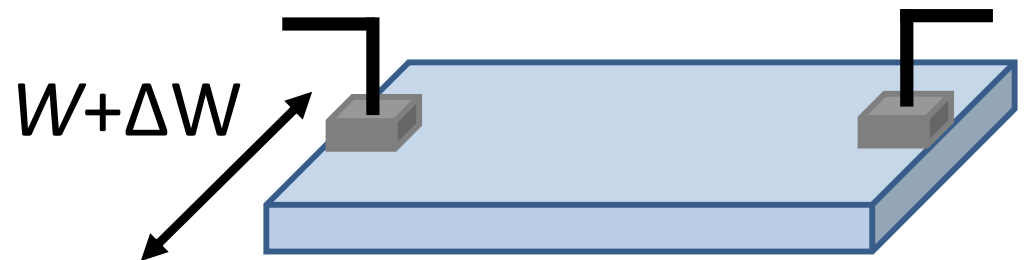
Ideal case



Real world

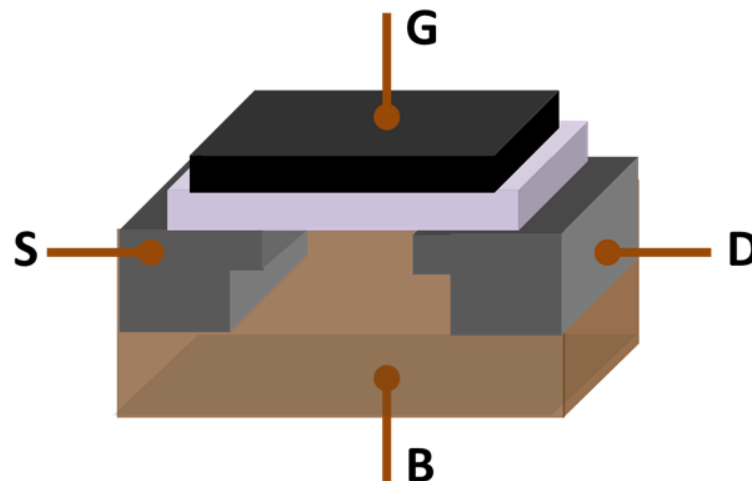
Spread of CMOS components

- For a resistor (tens of %)
 - Lithography error $\rightarrow W$ & L
 - Doping level variation $\rightarrow R_{\square}$ and TC



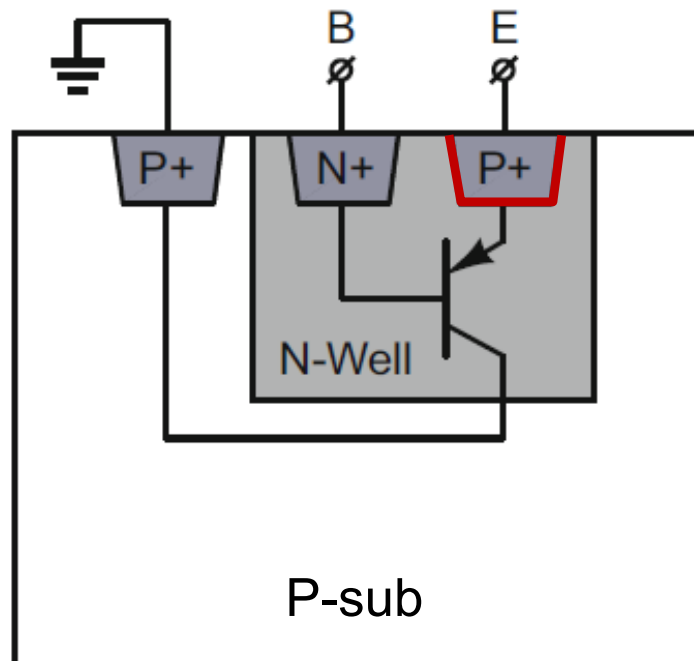
Spread of CMOS components

- For a MOSFET (tens of %)
 - Lithography error -> W & L
 - Doping level variation -> μ_n , V_{th} , etc.



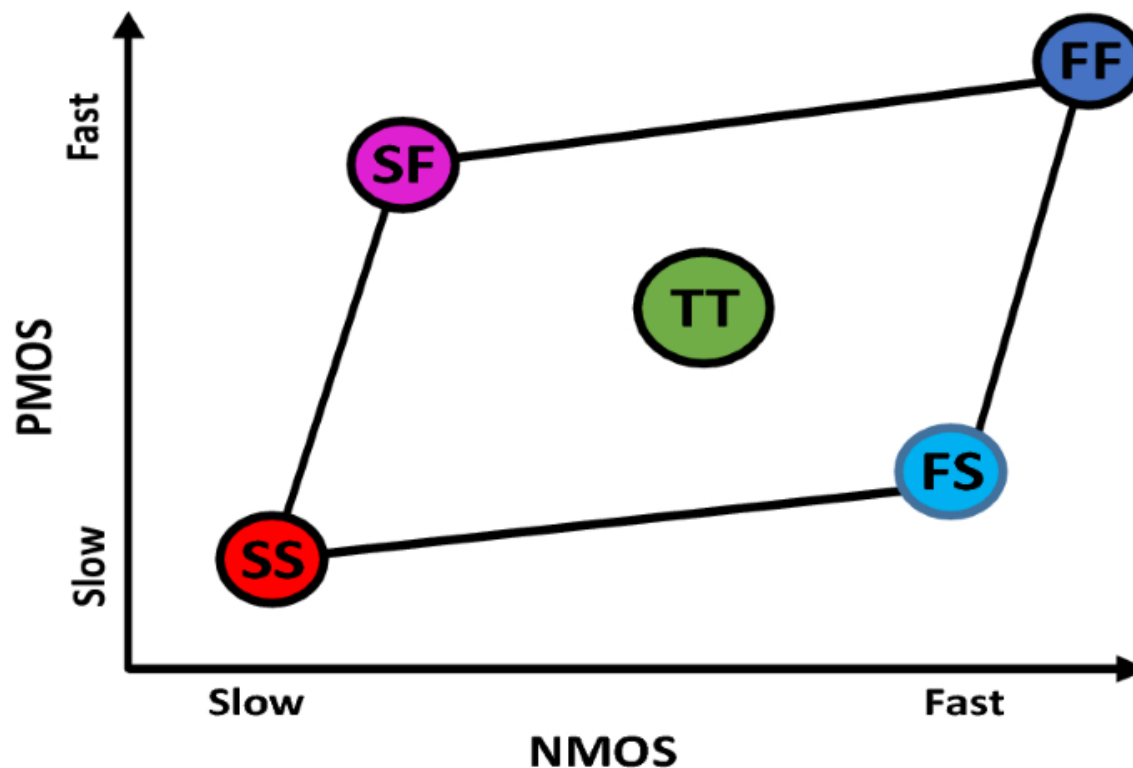
Spread of CMOS components

- For a BJT (few %)
 - Lithography error -> base-emitter area and R_B
 - Doping level variation -> I_S and β_F .



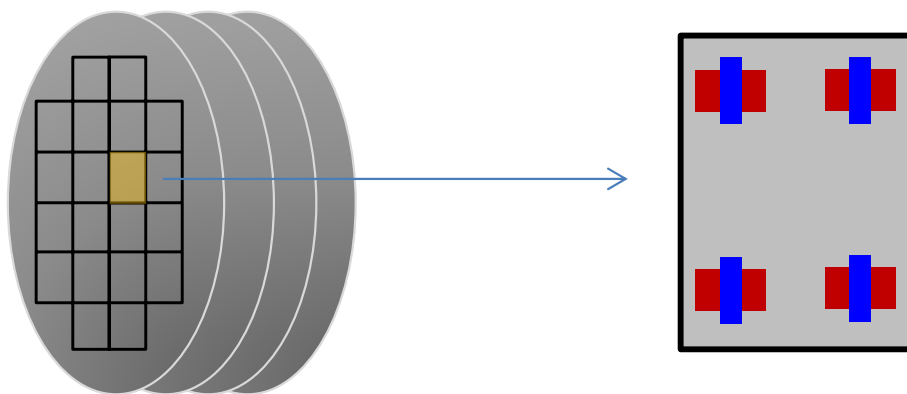
Spread of CMOS components

- Process corner:
 - The maximum spread from one batch to another



Spread of CMOS components

- Mismatch:
 - Spread among components on the same chip
 - Depends on size and layout, can be $< 0.1\%$ for “large” devices.

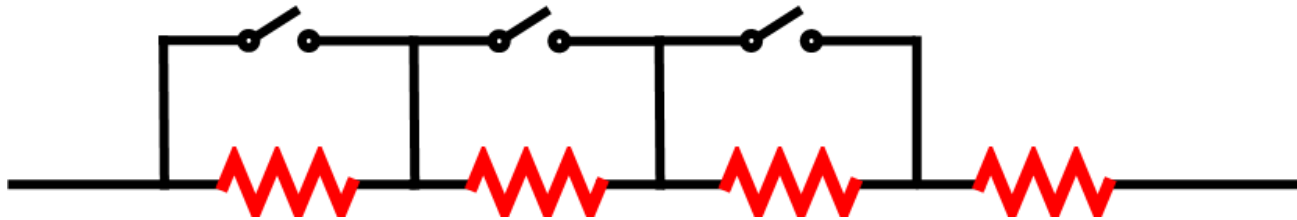


Design considerations

- Constant gm circuit:
 - Usually designed for circuit stability/noise
 - Not very critical, a 20% spread is often tolerable
- Bandgap and temperature sensors:
 - Targeting for a high accuracy ($\sim 1\%$)
 - Can be challenging

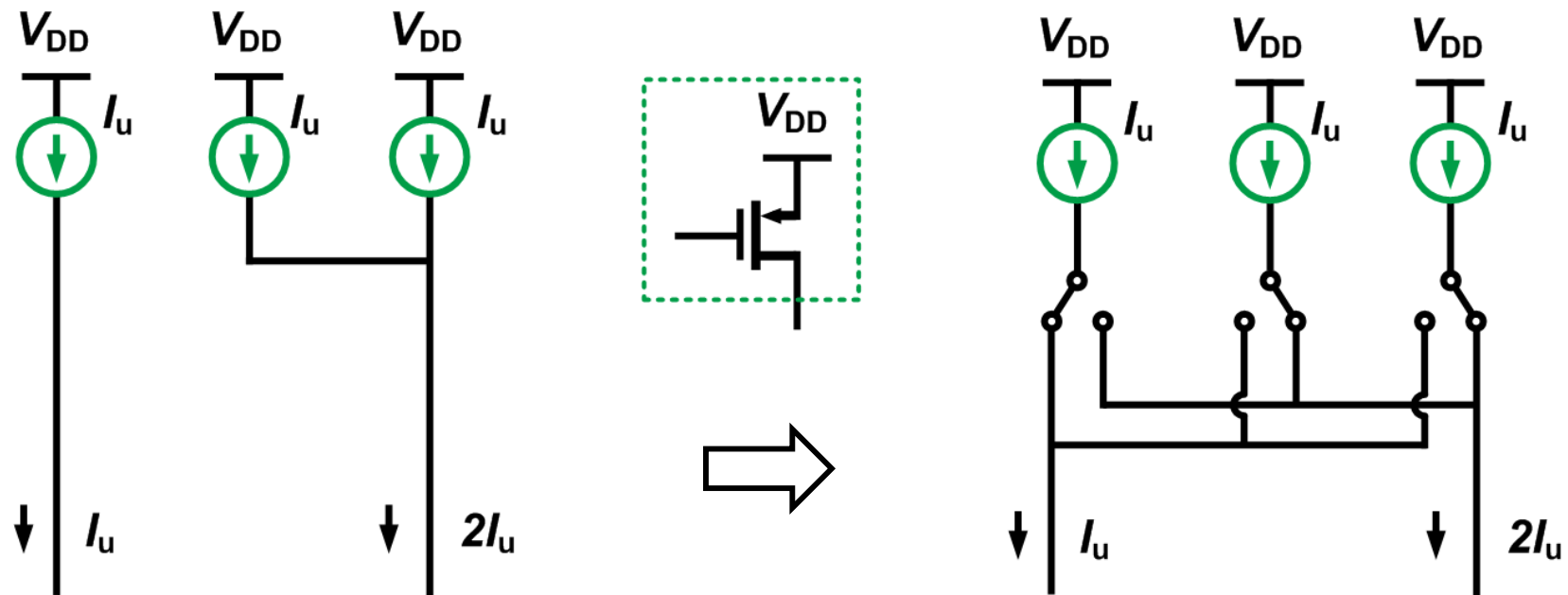
Coping with process spread

- Obtaining an accurate absolute value, e.g. a resistance or biasing current?
- => Trimming



Coping with mismatch

- Obtaining a fixed ratio, e.g. a current ratio.
- Dynamic element matching (DEM)
(EE4C08 Measurement and instrumentation)

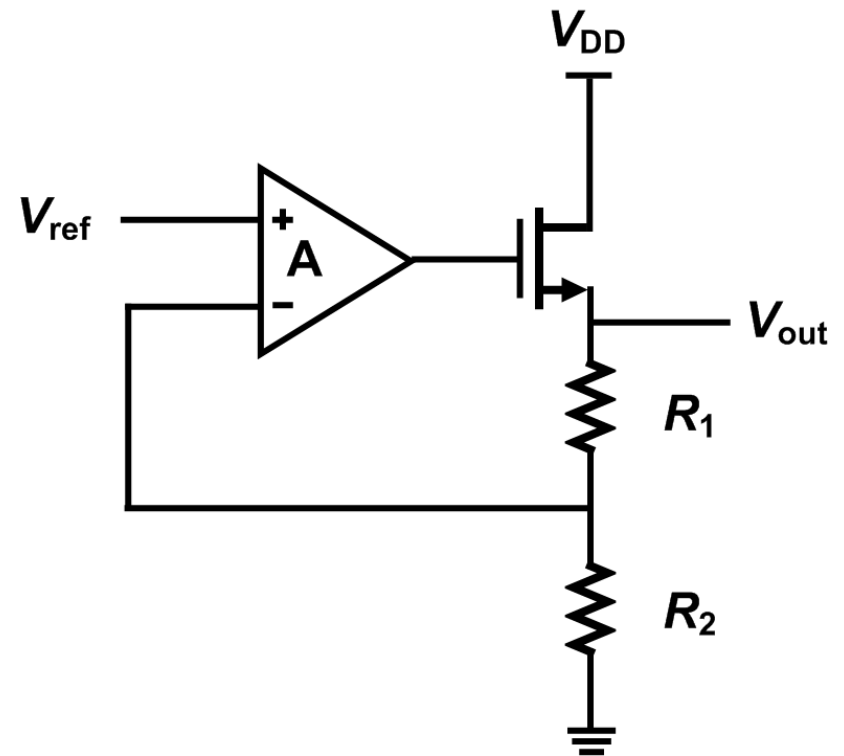


Bandgap reference: background

- Voltage regulator

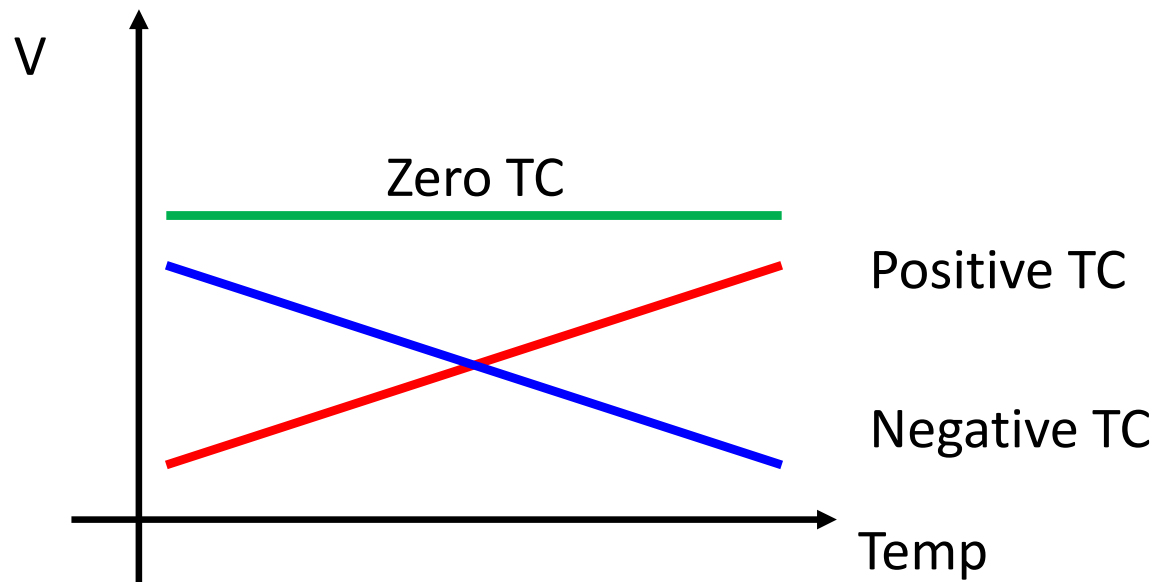
$$V_{out} = V_{ref} \cdot \frac{R_1 + R_2}{R_1}$$

- Constant V_{out} over PVT
-> requires a stable V_{ref}



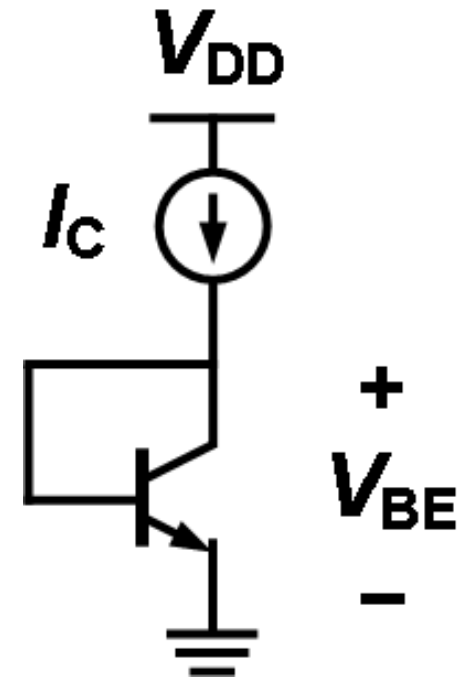
Bandgap reference principle

- Generate voltages with positive and negative temperature coefficients (TCs).
- Add them with proper weighting -> zero TC.



Negative-TC voltage

- For a bipolar device
- $I_C \approx I_S \cdot e^{\frac{V_{BE}}{V_T}}$
 - $I_S \approx bT^{2.5} e^{\frac{-E_g}{kT}}$
 - b is a const, $E_g \approx 1.12\text{eV}$
- Assuming a constant I_C ,
 $\frac{\partial V_{BE}}{\partial T} \approx -1.5\text{mV}/^\circ\text{C}$ with $V_{BE} \approx 750\text{mV}$
- Surprisingly linear over temperature!



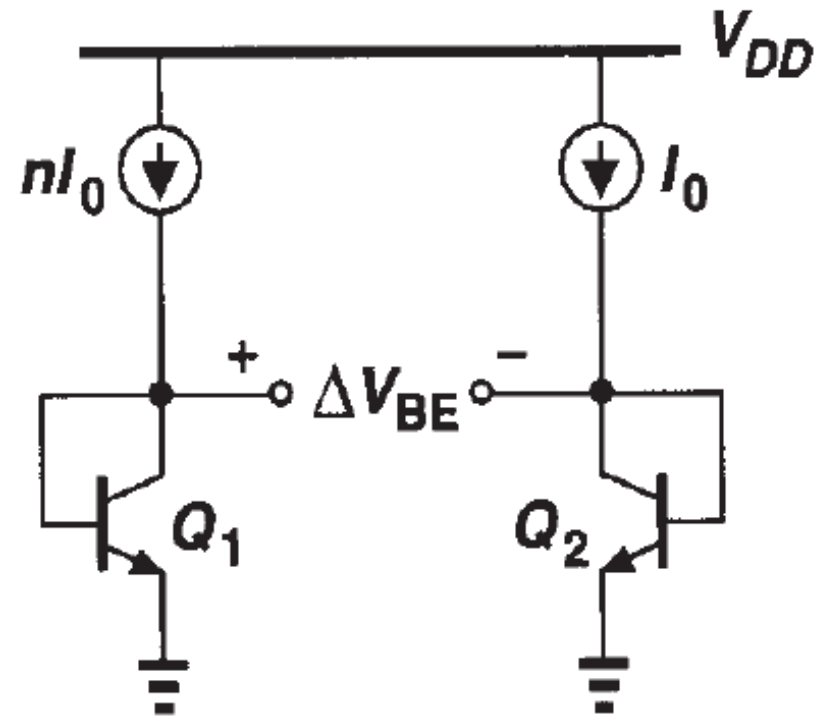
Positive-TC voltage

- Two identical transistors ($I_{S1} = I_{S2}$)

- $I_C \approx I_S \cdot e^{\frac{V_{BE}}{V_T}}$

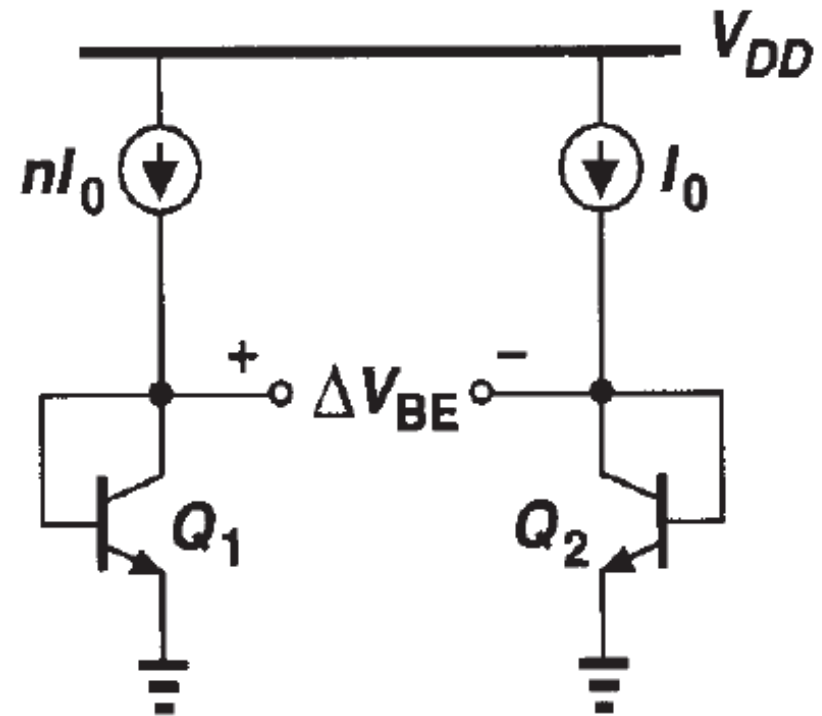
- $V_{BE1} = V_T \cdot \ln\left(\frac{nI_0}{I_{S1}}\right)$

- $V_{BE2} = V_T \cdot \ln\left(\frac{I_0}{I_{S2}}\right)$



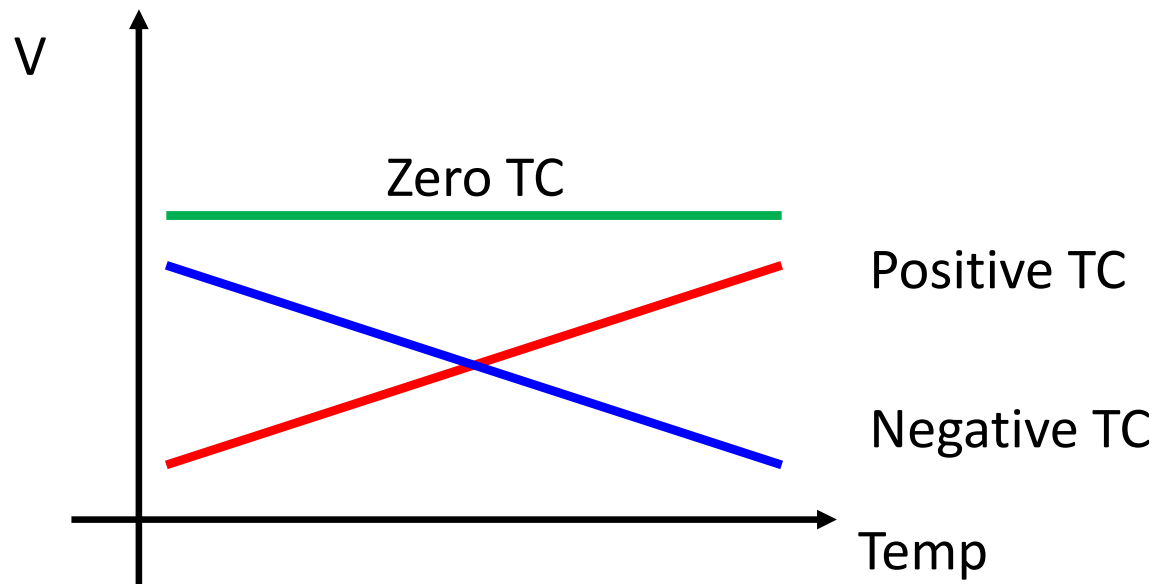
Positive-TC voltage

- Two identical transistors ($I_{S1} = I_{S2}$)
- $\Delta V_{BE} = V_T \left(\ln \frac{nI_0}{I_{S1}} - \ln \frac{I_0}{I_{S2}} \right)$
- $\Delta V_{BE} = V_T \cdot \ln n$
- $\frac{\partial \Delta V_{BE}}{\partial T} = \frac{k}{q} \cdot \ln n$
- $n=6 \rightarrow 0.15\text{mV}/^\circ\text{C}$



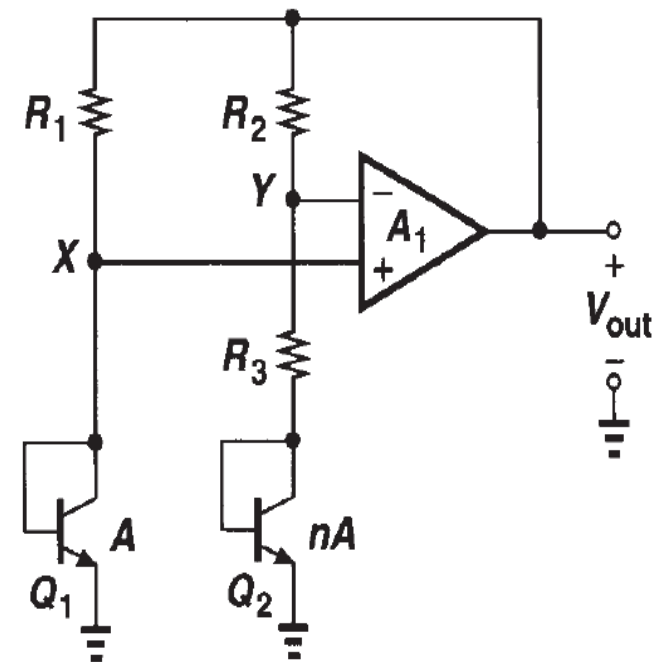
Bandgap reference principle

- $V_{REF} = \alpha_1 V_{BE} + \alpha_2 (V_T \cdot \ln n)$
- If $\alpha_1=1$, then $\alpha_2 \cdot \ln n = 17.2$
- $\rightarrow V_{REF} \approx V_{BE} + 17.2 V_T \approx 1.25V$



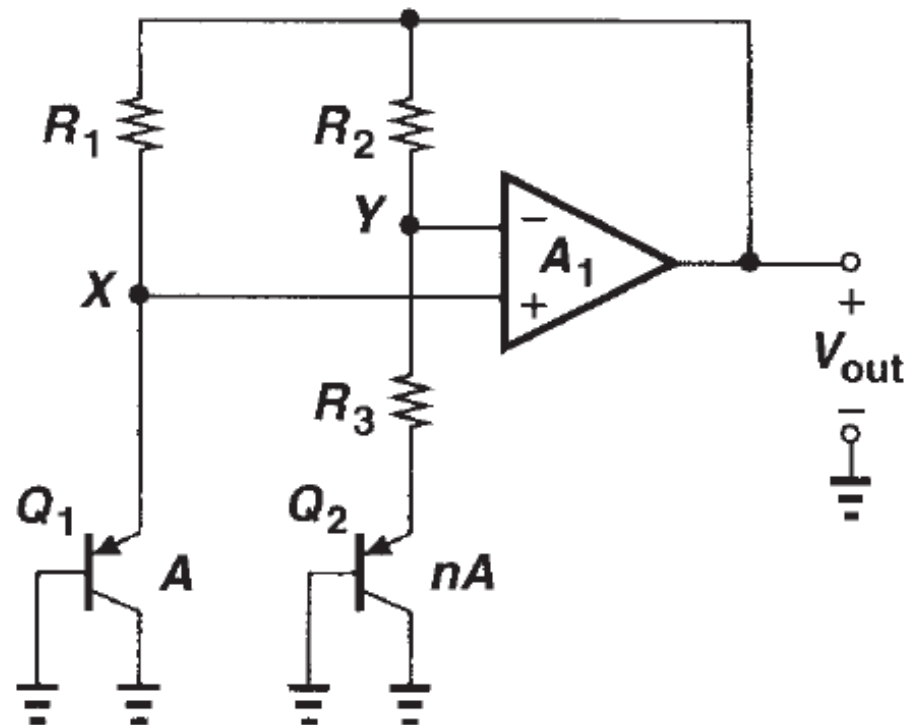
Actual implementation

- Area ratio= n , $R_1 = R_2$
- Opamp forces $V_X = V_Y$
- $I_{R3} = \Delta V_{BE}/R_3 = V_T \ln n / R_3$
- $V_{out} = V_{BE2} + V_T \ln n \cdot \left(1 + \frac{R_2}{R_3}\right)$
- $\rightarrow (1 + R_2/R_3) \ln n \approx 17.2$.
- E.g. $n = 31$ & $R_2/R_3 = 4$



CMOS compatibility

- Circuit implemented with PNP transistors
- Still there is $V_{out} = V_{BE2} + V_T \ln n \cdot \left(1 + \frac{R_2}{R_3}\right)$



Effect of spread

- $V_{out} = V_{BE2} + V_T \ln n \cdot \left(1 + \frac{R_2}{R_3}\right)$

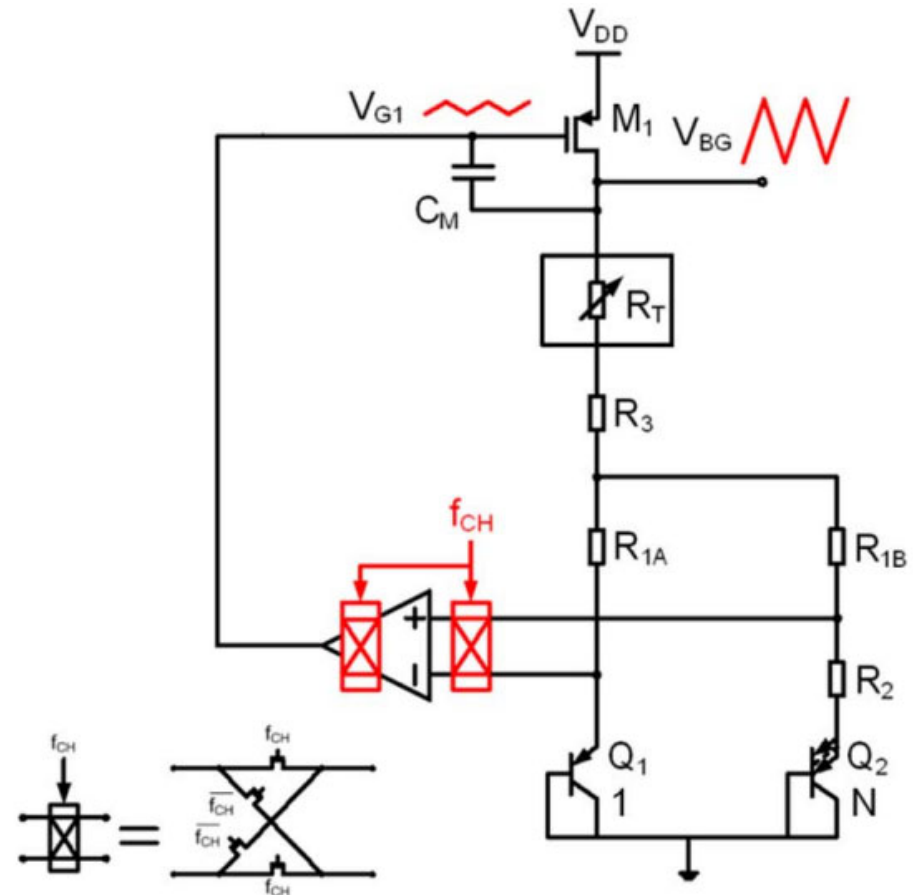
Dependent on I_S and
thus process spread

A ratio that can be
accurately obtained

- Result: V_{out} spread over different batches/samples, mainly due to the V_{BE} error.

State-of-the-art

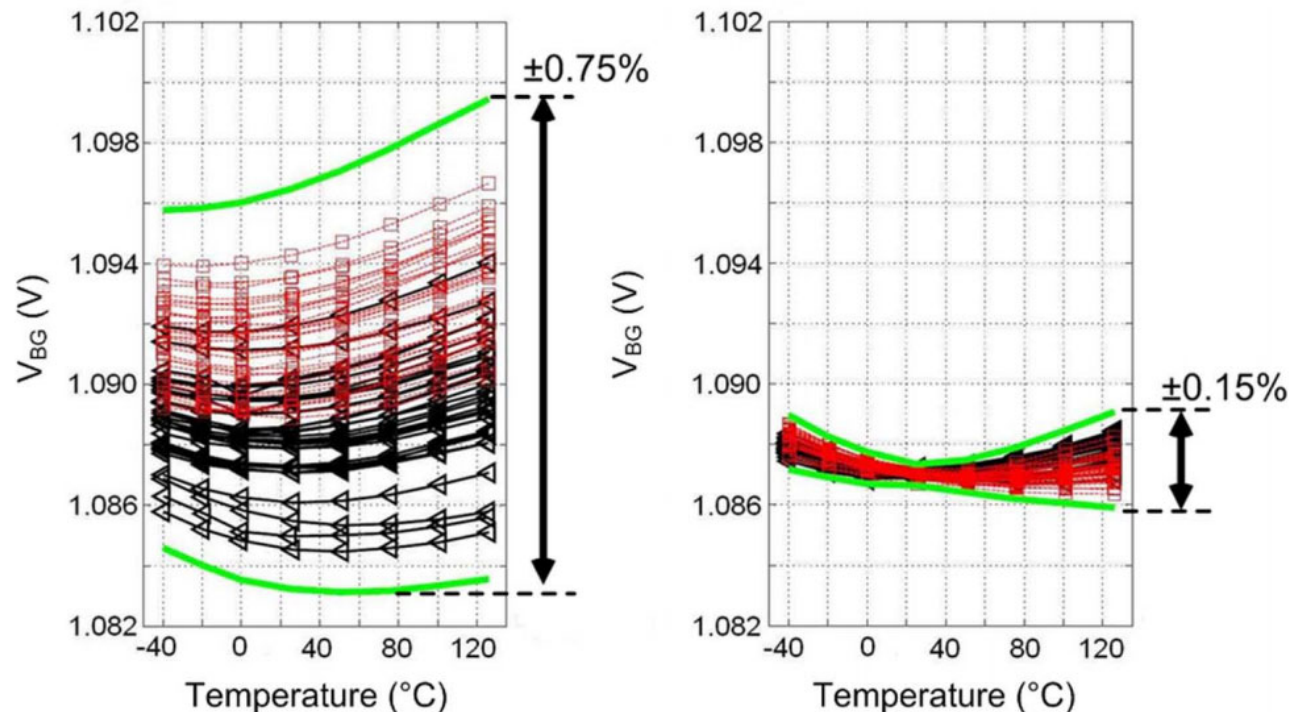
- I_S -related error removed by 1-pt trim
- Amplifier offset removed by chopping



Ge, Guang, et al. "A Single-Trim CMOS Bandgap Reference With a 3σ Inaccuracy of $\pm 0.15\%$ From -40°C to 125°C ." *IEEE Journal of Solid-State Circuits* 46.11 (2011): 2693-2701.

State-of-the-art

- 3σ Inaccuracy of $\pm 0.15\%$ from -40°C to 125°C



Ge, Guang, et al. "A Single-Trim CMOS Bandgap Reference With a 3σ Inaccuracy of $\pm 0.15\%$ From -40°C to 125°C ." *IEEE Journal of Solid-State Circuits* 46.11 (2011): 2693-2701.

The end!

- Practice session
- Oct 15th. 11:45 – 12:30
- Please download all the files from Brightspace in advance